Abir Hasan

Semiconductor devices | **IC design** | **Process technology**

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Summary

ECE Ph.D. candidate (expected May 2027) with hands-on experience across semiconductor devices, processes, and circuits. Built a theoretical model for oscillator-based computing hardware using nonlinear dynamical analysis, and analyzed MTJ devices with stochastic LLG simulations. Designed mixed-signal circuits in Spectre/HSPICE with process-corner and Monte Carlo analysis, through Calibre DRC/LVS/PEX sign-off and **tapeout** in **TSMC 65 nm**, and validated the chip against simulation. Fabricated 30 nm-wide nanowires on epitaxially grown 2D materials in a cleanroom using an **E-beam lithography system**, ICP-RIE dry-etch recipes, wet processing, and thin-film deposition, and characterized them electrically (resistivity, TCR, electromigration lifetime, contact resistivity).

Technical Skills

Device & Electrical Characterization: Probe-station DC I-V, resistivity scaling with width and temperature, Temperature Coefficient of Resistivity (TCR), electromigration lifetime, contact resistivity

Circuit Design & Simulation: Bandgap reference, ring oscillators, Schmitt trigger buffers, differential amplifiers, current mirrors, Pass-Transistor Logic; Monte Carlo variation analysis, process-corner verification

IC Implementation & Validation: Custom layout, DRC/LVS/PEX sign-off, post-layout extraction and tuning, test-bench development, PCB design (KiCad), post-silicon testing and debugging

EDA & Programming: Cadence Virtuoso, Spectre, ADE Explorer, HSPICE, Verilog, Verilog-A, Calibre, Cadence Innovus, MATLAB, Python

Process Technology: E-beam lithography, mask design, resist development, lift-off, ICP-RIE dry-etch recipe development, wet etching, RCA cleaning, E-beam evaporation, sputtering, atomic layer deposition

Metrology: SEM, Optical and Stylus Profilometer, AFM, Ellipsometer, EDS

Experience

Graduate Research Assistant

Aug. 2022 – Present

University of Virginia, Department of Electrical and Computer Engineering — *Charlottesville, VA*

Device Characterization and Modeling

- Characterized the fabricated devices on a probe station, measuring resistivity versus device width and across temperature and extracting a temperature coefficient of resistivity (TCR) better than copper.
- Evaluated interconnect reliability through electromigration lifetime under accelerated stress and contact resistivity measurements, with data analysis in MATLAB and Python.
- Modeled magnetic tunnel junction (**MTJ**) probabilistic bits with stochastic LLG (sLLG) simulations, including spin-orbit torque, and extracted switching-probability characteristics in MATLAB.

Circuit Design

- Completed **tapeout** of a functional mixed-signal IC in **TSMC 65 nm** technology (1.5 mm^2, 1.2 V supply, 17 coupled oscillators), from circuit-block design and verification to full-custom layout, Calibre DRC/LVS/PEX sign-off, and post-layout tuning against extracted parasitics.
- Designed and verified analog/mixed-signal blocks in Spectre and HSPICE with Verilog-A testbenches, process-corner and Monte Carlo analysis:
  - Designed a bandgap reference with multilevel bias generation and a Schmitt trigger buffer with tunable hysteresis.
  - Tuned a differential ring oscillator to 30 MHz; silicon frequency measured close to simulation.
  - Designed Pass-Transistor Logic programmable coupling for all 136 oscillator pairs, with 7 signed coupling-strength levels set by digital control bits.
- Carried out post-silicon bring-up, testing, and debugging on custom PCB test boards to validate the fabricated chip against simulation.
- Performed graduate coursework on solid state devices, semiconductor characterization, analog circuit design, control systems, and hardware computing.

- As capstone project in a graduate analog circuit design course, designed an amplifier targeting 5000 gain, 2 GHz bandwidth, 400 mV_{pp} output swing, 30 $pA/\sqrt{Hz}$ and 100 mW power budget.
- Built a theoretical model for oscillator-based computing hardware using nonlinear dynamical analysis, linking circuit-level behavior to accuracy and guiding design choices that improve hardware performance.

Process Technology

- Fabricated 30 nm-wide nanowires on epitaxially grown 2D materials using a **30 kV E-beam lithography system** (RAITH150 two) to benchmark against Cu BEOL baselines.
- Integrated the full device process flow (resist coating, E-beam exposure and development, metal deposition by E-beam evaporation or sputtering, lift-off, multi-layer alignment) and ran process-parameter sweeps to optimize it, checking results with SEM, AFM, profilometry, ellipsometry, and EDS.
- Developed chlorine-based ICP-RIE dry-etch recipes (Oxford PlasmaPro 100 Cobra) for nanoscale pattern transfer; also performed wet processing (resist development, RCA cleaning, wet etches) and atomic layer deposition.

Lecturer, Electrical and Electronic Engineering
Nov. 2021 – June 2022

University of Scholars — *Dhaka, Bangladesh*

- Delivered lectures on VLSI circuit design and developed assessment rubrics.

Projects

Highly scaled 2D metallic materials as interconnects

Intel — *Sep. 2022 – Aug. 2024*

- Fabricated, characterized, and evaluated 2D-material-based nanoscale structures as interconnect alternatives to Cu for back-end-of-line (BEOL) scaling.

Synchronized oscillations for computing

Department of War: MURI — *Aug. 2024 – Present*

- Implemented Oscillator Ising Machines (OIM) in a CMOS integrated circuit platform and demonstrated maximum-likelihood MIMO detection using coupled-oscillator computation.

Probabilistic-bit engines

National Science Foundation (NSF) — *Nov. 2025 – Present*

- Designed a generalized probabilistic computing scheme with noisy bistable-latch circuits as tunable multi-state sampling engines, and analyzed MTJ-based probabilistic bits with stochastic LLG simulations.

Education

University of Virginia — Charlottesville, VA

Ph.D. in Electrical and Computer Engineering - Advisor: Prof. Nikhil Shukla — *Aug. 2022 – May 2027 (Expected)*

Islamic University of Technology — Gazipur, Bangladesh

B.Sc. in Electrical and Electronic Engineering — *Jan. 2017 – Feb. 2021*

Publications & Talks

Hasan, A., Ekanayake, E.M., Lee, K., Camsari, K. and Shukla, N., 2026. *Breakdown of Gradient-Flow Dynamics in Oscillator Ising Machines from Harmonic Misalignment.* arXiv preprint arXiv:2605.17243. (Under review: Physical Review Letters, 2026)

Hasan, A. and Shukla, N., 2026. *How Physical Dynamics Shape the Properties of Ising Machines: Evaluating Oscillators vs. Bistable Latches as Ising Spins.* arXiv preprint arXiv:2603.06843. (Under review: Physical Review Applied, 2026)

Hasan, A. and Shukla, N., 2026. *At the top of the mountain, the world can look Boltzmann-like: Sampling dynamics of noisy double-well systems.* arXiv preprint arXiv:2602.04014. (Under review: Physical Review Applied, 2026)

Tonni, F.F., Maliat, M., Bati, S., Akhanda, M.S., Chandra, H., Scott, E.A., **Hasan, A.**, Krylyuk, S., Shukla, N., Constantin, C. and Hopkins, P.E., 2026. *Thermal transport in γ-InSe: Bulk single crystals and thin flakes.* Applied Physics Letters, 128(18).

Conference talk, *AVS 69th International Symposium and Exhibition*, Portland, OR — 2D materials as interconnects (Nov. 2023).